GIG Coordinate Evidence Card Template
P0 Architecture Document

Central template for coordinate-level evidence, source quality, confidence, risk, and decision output

Field	Definition
Document Role	Defines the standard operating template for each coordinate evaluated by GIG.
Project Positioning	Pre-Failure Coordinate Intelligence Layer.
Primary Output	Coordinate Evidence Card with evidence, source, confidence, risk, missing evidence, and decision status.
Decision States	Watch / Verify / Intervene / Escalate.
Use Mode	Standalone, or through adapters with HydroCode, White Petrol, and Oil Shale Green.
Status	Foundation architecture template for controlled review and pilot design.

1. Purpose of the Coordinate Evidence Card:

The Coordinate Evidence Card is the central operating unit of GIG. It converts a geographic coordinate from a point on a map into a structured, defensible evidence record that can support early verification, intervention prioritization, and controlled escalation before operational loss appears.
The card does not claim that a coordinate has failed, nor does it replace field inspection, environmental testing, geotechnical investigation, hydrological modeling, or specialist review. Its purpose is to organize what is known, what is missing, what is uncertain, and what decision is justified at the current evidence level.

2. Core Design Principles:

Code	Principle	Operational meaning
P1	Evidence before decision	No coordinate should be escalated without documented evidence, missing-evidence flags, and confidence boundaries.
P2	Source-linked claims	Every relevant observation must be connected to a source, source date, source type, and source reliability note.
P3	Uncertainty is explicit	Uncertainty is not hidden; it is recorded as a decision variable.
P4	Risk is not proof	Risk signals identify where verification or action may be required; they do not establish final causation.

GIG | Pre-Failure Coordinate Intelligence Layer | Controlled Disclosure Document | © 2026 Dana Obeidat. All Rights Reserved.

Code	Principle	Operational meaning
P5	Decision states are controlled	Each coordinate must resolve into Watch, Verify, Intervene, or Escalate.
P6	Human review remains required	GIG supports decisions; it does not automatically replace technical or institutional judgment.

3. Card Header: Coordinate Identity:

The card header establishes the identity of the coordinate and prevents ambiguity between similar points, assets, map layers, or field observations.

Field	Description	Entry guidance
Card ID	Unique card identifier	GIG-AREA-COORD-###
Coordinate ID	Internal coordinate reference	May be grid-based, asset-based, or project-based.
Latitude / Longitude	Primary coordinate values	Must include coordinate system and datum where known.
Coordinate System / Datum	Spatial reference basis	Required where maps or GIS layers are used.
Area / Zone / Corridor	Administrative or operational context	Example: road segment, water network zone, pilot site boundary.
Asset Link	Related infrastructure or natural feature	Pipe, road, culvert, drainage outlet, slope, site, borehole, etc.
Card Status	Draft / Under Review / Verified / Archived	Indicates maturity of the card, not risk severity.
Last Update	Most recent revision date	Required for audit trail and version control.

4. Source and Map Memory Section:

This section records how the coordinate memory is built from legacy maps, current datasets, field records, asset registers, inspection reports, complaints, satellite imagery, drainage information, environmental data, and any other approved source. The goal is not to accumulate data volume; the goal is to preserve data lineage and decision relevance.

Source ID	Source Name	Type	Date / Period	Spatial relevance	Reliability note
SRC-001	[Insert source]	Map / Satellite / Asset / Field / Lab / Report	[Date]	Direct / Nearby / Contextual	[Known limit, scale, completeness, confidence]
SRC-002	[Insert source]	[Type]	[Date]	[Relevance]	[Reliability note]
SRC-003	[Insert source]	[Type]	[Date]	[Relevance]	[Reliability note]
SRC-004	[Insert source]	[Type]	[Date]	[Relevance]	[Reliability note]

Rule:

A source may support a signal, but it must not be treated as final proof unless the source is designed for that purpose and has acceptable quality boundaries.

5. Evidence Inventory Section:

The evidence inventory separates observations from interpretation. Each item records what was observed, where it came from, what it may indicate, and what it cannot prove by itself.

Evidence ID	Observation	Source	Evidence layer	Strength	Quality	Boundary
EV-001	[Observed item]	[Source ID]	Ground / Water / Drainage / Asset / Map / Complaint / Lab / Other	Direct / Indirect / Contextual	Low / Medium / High	[Limit or uncertainty]
EV-002	[Observed item]	[Source ID]	[Layer]	[Evidence strength]	[Quality]	[Limit]
EV-003	[Observed item]	[Source ID]	[Layer]	[Evidence strength]	[Quality]	[Limit]

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Evidence ID	Observation	Source	Evidence layer	Strength	Quality	Boundary
EV-004	[Observed item]	[Source ID]	[Layer]	[Evidence strength]	[Quality]	[Limit]

6. Risk Signal Section:

Risk signals describe what may be accumulating at the coordinate. They must be treated as indicators requiring review, not as automatic conclusions. Signals may come from ground behavior, drainage exposure, water pathway sensitivity, asset exposure, missing documentation, historical disturbance, or operational anomalies.

Risk signal	Signal level	Supporting evidence	Boundary / caution
Ground / Soil Sensitivity	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Drainage / Runoff Exposure	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Water Pathway Sensitivity	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Asset Exposure	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Maintenance / Complaint Pattern	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Historical Disturbance / Legacy Activity	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[Interpretation boundary]
Missing Evidence Risk	[None / Low / Medium / High / Unknown]	[Evidence IDs]	[What is missing]

7. Confidence and Uncertainty Section:

Confidence and uncertainty must be recorded separately. A coordinate can have high risk indicators but low confidence, or moderate risk indicators with strong supporting evidence. GIG should not force a false certainty where the evidence is incomplete.

Confidence dimension	Level	Assessment guidance
Evidence Coverage	[Low / Medium / High]	How much of the required evidence exists for this coordinate.
Source Reliability	[Low / Medium / High]	Quality, relevance, age, and traceability of sources.
Spatial Confidence	[Low / Medium / High]	Confidence in coordinate accuracy, datum, map alignment, and spatial resolution.
Temporal Confidence	[Low / Medium / High]	Whether the evidence is recent enough or historically relevant.
Interpretation Confidence	[Low / Medium / High]	Whether the observed signals support a reasonable interpretation.
Overall Uncertainty Level	[Low / Medium / High / Critical]	Combined uncertainty requiring caution, verification, or escalation.

8. Decision Output Section:

The card must resolve into one controlled decision state. The decision state should be based on the relationship between risk signals, evidence quality, missing evidence, asset exposure, and the consequence of delay.

Decision state	When used	Typical next action
Watch	Signals are present but do not justify immediate verification or intervention.	Schedule monitoring, preserve evidence, update after new data.
Verify	Risk is plausible but evidence is incomplete or uncertain.	Request field check, inspection, sampling, survey, or expert review.
Intervene	Evidence supports a limited practical action within existing authority.	Issue maintenance, drainage clearing, localized inspection, asset protection, or controlled mitigation.
Escalate	Potential consequence, uncertainty, or institutional exposure exceeds routine action.	Raise to decision authority, budget review, emergency review, or third-party verification.

9. Recommended Action and Work Order Link:

Where possible, the card should connect the coordinate to an operational action. This prevents GIG from becoming a passive dashboard and keeps the system aligned with its purpose: earlier, defensible intervention logic.

Action field	Entry
Recommended next action	[Insert specific action]
Responsible function / owner	[Water authority / roads / municipality / pilot team / technical reviewer / other]
Required verification	[Field inspection / sample / survey / source check / engineering review / none]
Priority level	[Low / Medium / High / Critical]
Target timeframe	[Immediate / 7 days / 30 days / next update cycle / other]
Work Order / Review ID	[Insert ID if available]
Decision rationale	[Why this decision state is justified based on evidence and uncertainty]

10. Adapter Context Section:

The same card must remain usable when GIG operates standalone or through a project adapter. The adapter should not change the evidence discipline; it only changes the operational lens.

Mode	Adapter lens	Card relevance
Standalone GIG	Infrastructure, water, drainage, ground, and asset risk intelligence.	Coordinate risk and decision readiness.
HydroCode Adapter	Water system truth, pressure, flow, quality, anomaly context.	Water-ground risk linkage and upstream/downstream review.
White Petrol Adapter	Pilot site readiness, environmental memory, inherited site risk.	Avoid entering a pilot site without evidence readiness.
Oil Shale Green Adapter	Mining legacy, ground memory, groundwater sensitivity, reclamation context.	Separate inherited ground risk from new project impact.

11. Full Blank Card Template:

The following blank format may be copied for each coordinate in a pilot or internal review. It is intentionally structured to support repeatability, auditability, and controlled decision-making.

Card section	Template entry
A. Coordinate Identity	Card ID: ___ Coordinate ID: ___ Latitude/Longitude: ___ Datum/Coordinate System: ___ Area/Zone: ___ Linked Asset/Site: ___ Card Status: ___ Last Update: ___
B. Source Register Snapshot	Primary sources: ___ Source dates: ___ Source types: ___ Source quality notes: ___ Known gaps: ___
C. Map Memory Summary	Legacy map evidence: ___ Current map/data evidence: ___ Conflicting records: ___ Missing map or metadata items: ___
D. Evidence Inventory	Evidence IDs: ___ Observations: ___ Supporting sources: ___ Evidence boundaries: ___
E. Risk Signals	Ground/soil: ___ Drainage/runoff: ___ Water pathway: ___ Asset exposure: ___ Maintenance/complaints: ___ Historical disturbance: ___ Missing evidence: ___
F. Confidence and Uncertainty	Evidence coverage: ___ Source reliability: ___ Spatial confidence: ___ Temporal confidence: ___ Interpretation confidence: ___ Overall uncertainty: ___
G. Decision Output	Decision state: Watch / Verify / Intervene / Escalate Decision rationale: ___ Recommended next action: ___

Card section	Template entry
	Owner: ___ Target timeframe: ___
H. Verification and Audit Trail	Verification required: ___ Reviewer: ___ Review date: ___ Work order / review ID: ___ Revision notes: ___

12. Use Rules and Quality Gates:

- A Coordinate Evidence Card must not be marked as verified unless the required evidence has been reviewed or the verification limitation is explicitly documented.
- A high risk score must not be interpreted as proof of failure; it is a prioritization signal for review or action.
- A missing evidence flag must remain visible until the missing source, inspection, sample, or review is completed or formally waived.
- The card must preserve source identity, evidence boundaries, and uncertainty labels across all adapters.
- No adapter may remove the Watch / Verify / Intervene / Escalate decision discipline.
- Any field action triggered by the card should be linked to a responsible owner, date, and work order or review reference where available.

13. Position within the GIG Architecture:

The Coordinate Evidence Card sits between Data Foundation / Map Memory and the Risk Intelligence Engine. It is the structured handoff point where raw data becomes organized evidence before risk scoring, uncertainty handling, and decision output.

Step	Architecture layer	Role
1	Data Foundation	Collect and register maps, legacy records, current datasets, asset information, field evidence, and source metadata.
2	Map Memory	Build coordinate memory by preserving historical and current spatial context.
3	Coordinate Evidence Card	Convert coordinate memory into evidence, confidence, risk signals, and decision readiness.
4	Risk Intelligence Engine	Rank risk, uncertainty, missing evidence, and asset exposure.
5	Decision & Work Order Engine	Produce Watch / Verify / Intervene / Escalate and connect to action.

14. Closing Statement:

The Coordinate Evidence Card is the practical center of GIG. It prevents the project from becoming a map, a dashboard, or an unchecked prediction system. It forces every coordinate to carry its evidence, source quality, missing evidence, uncertainty, risk signal, and decision status in one defensible structure.

In the first Minimum Defensible Pilot, this template should be applied to a controlled set of 50-100 coordinates using 5-7 approved data sources. The pilot should test not whether GIG predicts all failures, but whether it improves early, defensible decision readiness at coordinate level.